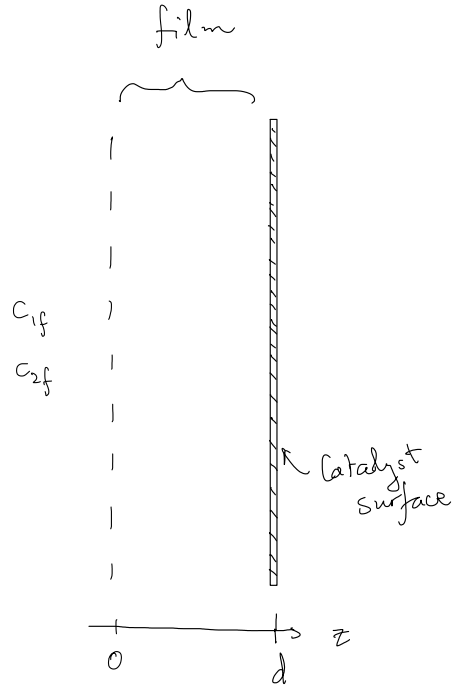
Q1] $A_1 + A_2 \rightarrow 2A_3$; $\hat{r}_s = \hat{k} C_{1s} C_{2s}$ (Conc. @ surface) ; ideal gas

$A_1 + A_2 - 2A_3 = 0$

$j = 1, 2, 3$

(i) 1D diffusion through a film — Bulk conc.: C_{1f}, C_{2f}

From notes: $\frac{dx_i}{dz} = \sum_{j=1}^{S} \frac{1}{cD_{ij,e}} (x_i N_j - x_j N_i)$; $x_i = \frac{c_i}{c_T}$; @ constant T and P



$i=1$: $\frac{dx_1}{dz} = \frac{1}{cD_{11,e}}(x_1 N_1 - x_1 N_1) + \frac{1}{cD_{12,e}}(x_1 N_2 - x_2 N_1) + \frac{1}{cD_{13,e}}(x_1 N_3 - x_3 N_1)$

$\frac{dx_1}{dz} = \frac{1}{D_{11,e}}(x_1 N_1 - x_1 N_1) + \frac{1}{D_{12,e}}(x_1 N_2 - x_2 N_1) + \frac{1}{D_{13,e}}(x_1 N_3 - x_3 N_1)$

$$c \frac{dc_1}{dz} = \frac{1}{D_{12,e}}(c_1 N_2 - c_2 N_1) + \frac{1}{D_{13,e}}(c_1 N_3 - c_3 N_1)$$

$i=2$: $\frac{dx_2}{dz} = \frac{1}{cD_{21,e}}(x_2 N_1 - x_1 N_2) + \frac{1}{cD_{22,e}}(x_2 N_2 - x_2 N_2) + \frac{1}{cD_{23,e}}(x_2 N_3 - x_3 N_2)$

Similarly,

$$c \frac{dc_2}{dz} = \frac{1}{D_{21,e}}(c_2 N_1 - c_1 N_2) + \frac{1}{D_{23,e}}(c_2 N_3 - c_3 N_2)$$

$i=3$: $\frac{dx_3}{dz} = \frac{1}{cD_{31,e}}(x_3 N_1 - x_1 N_3) + \frac{1}{cD_{32,e}}(x_3 N_2 - x_2 N_3) + \frac{1}{cD_{33,e}}(x_3 N_3 - x_3 N_3)$

$$c \frac{dc_3}{dz} = \frac{1}{D_{31,e}}(c_3 N_1 - c_1 N_3) + \frac{1}{D_{32,e}}(c_3 N_2 - c_2 N_3)$$

(ii) Continuity eq: $\frac{dN_j}{dz} = \alpha_j r$; Since no reaction is happening in the film, $r = 0$

@ $z = 0$: $\frac{dN_1}{dz} = 1r = 0$; $\frac{dN_2}{dz} = 1r = 0$; $\frac{dN_3}{dz} = -2r = 0$ } Constant flux in the film for N_1, N_2, N_3 ∆ conc. is equal to bulk concentration

@ $z = \delta$: $N_1 = \alpha_1 \hat{r}$; $N_2 = \alpha_2 \hat{r}$; $N_3 = \alpha_3 \hat{r}$; $\hat{r}_s = \hat{k}_s C_{1s} C_{2s}$

$\alpha_1 = -1$; $\alpha_2 = -1$; $\alpha_3 = 2$

$$-N_1 = -N_2 = \frac{N_3}{2} = \hat{r}_s$$

Plugging in the (1) equation:

$$c \frac{dc_1}{dz} = \frac{1}{D_{12,e}}(-c_1(\hat{r}_s) + c_2 \hat{r}_s) + \frac{1}{D_{13,e}}(2\hat{r}_s c_1 + c_3 \hat{r}_s) = \frac{\hat{r}_s}{D_{12,e}}(-c_1 + c_2) + \frac{\hat{r}_s}{D_{13,e}}(2c_1 + c_3)$$

$$c \frac{dc_2}{dz} = \frac{1}{D_{21,e}}(-c_2 \hat{r}_s + c_1 \hat{r}_s) + \frac{1}{D_{23,e}}(2c_2 \hat{r}_s + c_3 \hat{r}_s) = \frac{\hat{r}_s}{D_{21,e}}(-c_2 + c_1) + \frac{\hat{r}_s}{D_{23,e}}(2c_2 + c_3)$$

$$c \frac{dc_3}{dz} = \frac{1}{D_{31,e}}(c_3(-\hat{r}_s) - 2\hat{r}_s c_1) + \frac{1}{D_{32,e}}(-c_3 \hat{r}_s - 2c_2 \hat{r}_s) = \frac{\hat{r}_s}{D_{31,e}}(-c_3 - 2c_1) + \frac{\hat{r}_s}{D_{32,e}}(-c_3 - 2c_2)$$

For ideal gas : $\frac{N}{V} = c = \frac{RT}{P}$; $\frac{dc}{dz} = \frac{d\left(\frac{RT}{P}\right)}{dz} = 0$

$a\pm$ $\quad \nu_{31,e}$ $\quad \nu_{32,e}$ $\quad \nu_{33,e}$

For ideal gas : $\frac{N}{V} = c = \frac{RT}{P}$; $\frac{dc}{dz} = \frac{d\left(\frac{RT}{P}\right)}{dz} = 0$

Total conc. is conserved ; Total moles are conserved
(No variation in z axis)

(iii) For non-porous catalysts , Effectiveness factor is defined as actual reaction rate (including diffusion) vs the reaction rate when entire surface is at bulk conditions

$$\therefore \eta = \frac{\int_A \hat{r}_s \, dS}{A \cdot \hat{r}_f} \quad ; \quad h = \frac{\hat{r}_s}{\hat{r}_f} = \eta$$

(iv) Non-dimensionalizing; $\phi^2 = \frac{\hat{r}_f \delta}{c D_{12}}$; $\zeta = \frac{z}{\delta}$; $\hat{r}_f = \hat{k} c_{1f} c_{2f}$; $\hat{r}_s = h \hat{r}_f$

$$\frac{dc_1}{dz} = \frac{\hat{r}_s}{c D_{12,e}} (-c_1 + c_2) + \frac{\hat{r}_s}{c D_{13,e}} (2c_1 + c_3)$$

$$\frac{dc_1}{d(\zeta\delta)} = \frac{h\hat{r}_f}{c D_{12,e}} (-c_1 + c_2) + \frac{h\hat{r}_f}{c D_{13,e}} (2c_1 + c_3)$$

$$\frac{dc_1}{d\zeta} = \frac{h\hat{r}_f \delta}{c D_{12,e}} (-c_1 + c_2) + \frac{h\hat{r}_f \delta}{c D_{n,e}} \frac{D_{12,e}}{D_{13,e}} (2c_1 + c_3)$$

$$\boxed{\frac{dc_1}{d\zeta} = h\phi^2 (-c_1 + c_2) + h\phi^2 \frac{D_{12,e}}{D_{13,e}} (2c_1 + c_3)}$$

$$\frac{dc_2}{d(\zeta\delta)} = \frac{h\hat{r}_f}{c D_{12,e}} \frac{D_{12,e}}{D_{21,e}} (-c_2 + c_1) + \frac{h\hat{r}_f}{c D_{12,e}} \cdot \frac{D_{12,e}}{D_{23,e}} (2c_2 + c_3)$$

$$\boxed{\frac{dc_2}{d\zeta} = h\phi^2 \frac{D_{12,e}}{D_{21,e}} (-c_2 + c_1) + h\phi^2 \frac{D_{12,e}}{D_{23,e}} (2c_2 + c_3)}$$

$$\frac{dc_3}{d(\zeta\delta)} = \frac{h\hat{r}_f}{c D_{12,e}} \frac{D_{12,e}}{D_{31,e}} (-c_3 - 2c_1) + \frac{h\hat{r}_f}{c D_{13,e}} \frac{D_{12,e}}{D_{32,e}} (-c_3 - 2c_2)$$

$$\boxed{\frac{dc_3}{d\zeta} = h\phi^2 \frac{D_{12,e}}{D_{31,e}} (-c_3 - 2c_1) + h\phi^2 \frac{D_{12,e}}{D_{32,e}} (-c_3 - 2c_2)}$$

BC's:

@ $\zeta = 0$: $c_1 = c_{1f}$, $c_2 = c_{2f}$ (bulk concentrations)

@ $\zeta = 1$: Diffusion rate = reaction rate

$$N_1 = -D_{1,e} \frac{dc_1}{dz} = -\hat{r}_s$$

$$N_2 = -D_{2,e} \frac{dc_2}{dz} = -\hat{r}_s$$

$$N_3 = -D_{3,e} \frac{dc_3}{dz} = 2\hat{r}_s$$

(v) Simplifying the expression using $1 = \sum_i x_i$; $c_i = x_i c$

$$\frac{dc_1}{d\tau_e} = h\phi^2\left(-c_1 + c_2\right) + h\phi^2 \frac{D_{12,e}}{D_{13,e}}\left(2c_1 + c_3\right)$$

$$\frac{dx_1}{d\tau_e} = h\phi^2\left(-x_1 + x_2\right) + h\phi^2 \frac{D_{12}}{D_{13}}\left(2x_1 + x_3\right)$$

$$\frac{dx_1}{d\tau_e} = h\phi^2\left(-x_1 + x_2\right) + h\phi^2 \frac{D_{12}}{D_{13}}\left(2x_1 + (1 - x_1 - x_2)\right)$$

$$\frac{dx_1}{d\tau_e} = h\phi^2\left(-x_1 + x_2\right) + h\phi^2 \frac{D_{12}}{D_{13}}\left(1 + x_1 - x_2\right)$$

$$\frac{dx_1}{d\tau_e} = h\phi^2\left[x_1\left(-1 + \frac{D_{12}}{D_{13}}\right) + x_2\left(1 - \frac{D_{12}}{D_{13}}\right) + \frac{D_{12}}{D_{13}}\right]$$

Similarly, $\frac{dc_2}{d\tau_e} = h\phi^2 \frac{D_{12,e}}{D_{21,e}}\left(-c_2 + c_1\right) + h\phi^2 \frac{D_{12,e}}{D_{23,e}}\left(2c_2 + c_3\right)$ $\quad 1 - x_2 - x_1$

$$\frac{dx_2}{d\tau_e} = h\phi^2\left[\frac{D_{12}}{D_{21}}\left(-x_2 + x_1\right) + h\phi^2 \frac{D_{12}}{D_{23}}\left(1 - x_1 + x_2\right)\right]$$

$$\frac{dx_2}{d\tau_e} = h\phi^2\left[\left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right)x_1 + \left(-\frac{D_{12}}{D_{21}} + \frac{D_{12}}{D_{23}}\right)x_2 + \frac{D_{12}}{D_{23}}\right]$$

Combining these as an array:

$$h^{-1}\phi^{-2}\frac{d}{d\tau_e}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \left(-1 + \frac{D_{12}}{D_{13}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(-\frac{D_{12}}{D_{21}} + \frac{D_{12}}{D_{13}}\right) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} D_{12}/D_{13} \\ D_{12}/D_{23} \end{bmatrix}$$

$\underline{x}$; $\underline{\underline{A}}$; $\underline{x}$; $\underline{b}$

(vi) Show given $\underline{x}$ is a solution:

LHS: $\underline{x}(\tau_e) = e^{n\phi^2 \underline{\underline{A}} \tau_e}\, \underline{x}_f + \underbrace{n\phi^2 \int_0^{\tau_e} e^{n\phi^2 \underline{\underline{A}} \tau'} \underline{b}\, d\tau'}$

Evaluating this

$$\Rightarrow \left(n\phi^2\right)\left(n\phi^2 \underline{\underline{A}}\right)^{-1} e^{n\phi^2 \underline{\underline{A}} \tau'} \Big|_0^{\tau_e} \underline{b}$$

$$\Rightarrow \underline{\underline{A}}^{-1}\left(e^{n\phi^2 \underline{\underline{A}} \tau_e} - \underline{\underline{I}}\right) \underline{b}$$

$$\underline{x}(\tau_e) = e^{n\phi^2 \underline{\underline{A}} \tau_e}\, \underline{x}_f + \underline{\underline{A}}^{-1}\left(e^{n\phi^2 \underline{\underline{A}} \tau_e} - \underline{\underline{I}}\right) \underline{b}$$

$$\frac{d\underline{x}}{d\tau_e} = n\phi^2 \underline{\underline{A}}\, e^{n\phi^2 \underline{\underline{A}} \tau_e}\, \underline{x}_f + \underline{\underline{A}}^{-1} n\phi^2 \underline{\underline{A}}\, e^{n\phi^2 \underline{\underline{A}} \tau_e} \underline{b}$$

$$= n\phi^2 e^{n\phi^2 \underline{\underline{A}} \tau_e}\left(\underline{\underline{A}}\, \underline{x}_f + \underline{b}\right)$$

$h^{-1}\phi^{-2}$

ala

$$= n\phi^2 e^{n\phi^2 \underline{A} t_a} \left(\underline{A}\, \underline{x}_f + \underline{b} \right)$$

$n^{-1}\phi^{-2} \times$

$$n^{-1}\phi^{-2} \frac{d\underline{x}}{dt_a} = e^{n\phi^2 \underline{A} t_a} \left(\underline{A}\, \underline{x}_f + \underline{b} \right)$$

Solving for RHS: $\underline{A}\underline{x} + \underline{b} \Rightarrow \underline{A}\left(e^{n\phi^2 \underline{A} t_a} \underline{x}_f + \underline{A}^{-1}\left(e^{n\phi^2 \underline{A} t_a} - \underline{I} \right)\underline{b} \right) + \underline{b}$

$$\Rightarrow \underline{A}\, e^{n\phi^2 \underline{A} t_a} \underline{x}_f + e^{n\phi^2 \underline{A} t_a} \underline{b} - \cancel{\underline{b}} + \cancel{\underline{b}}$$

$$\Rightarrow e^{n\phi^2 \underline{A} t_a} \left(\underline{A}\, \underline{x}_f + \underline{b} \right)$$

LHS = RHS ✓

$\underline{x}$ is a solution

(vii) Finding eigenvalues of $\underline{A} = \begin{bmatrix} \left(-1 + \frac{D_{12}}{D_{13}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(\frac{-D_{12}}{D_{21}} + \frac{D_{12}}{D_{23}}\right) \end{bmatrix}$

$$\det(\underline{A} - \lambda \underline{I}) = \left(-1 + \frac{D_{12}}{D_{13}} - \lambda\right)\left[\frac{-D_{12}}{D_{21}} + \frac{D_{12}}{D_{23}} - \lambda\right] - \left(1 - \frac{D_{12}}{D_{13}}\right)\left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) = 0$$

$$= \cancel{\frac{D_{12}}{D_{21}}} - \cancel{\frac{D_{12}}{D_{23}}} + \lambda - \cancel{\frac{D_{12}^2}{D_{21} D_{13}}} + \cancel{\frac{D_{12}^2}{D_{13} D_{23}}} - \lambda \frac{D_{12}}{D_{13}} + \lambda \frac{D_{12}}{D_{21}} - \lambda \frac{D_{12}}{D_{23}} + \lambda^2 - \left[\cancel{\frac{D_{12}}{D_{21}}} - \cancel{\frac{D_{12}}{D_{23}}} - \cancel{\frac{D_{12}^2}{D_{13} D_{21}}} + \cancel{\frac{D_{12}^2}{D_{13} D_{23}}} \right]$$

$$= \lambda \left(1 - \frac{D_{12}}{D_{13}} + \frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}} + \lambda \right) = 0$$

If $D_{21} = D_{12}$, $\boxed{\lambda = 0}$

$$1 - \frac{D_{12}}{D_{13}} + 1 - \frac{D_{12}}{D_{23}} + \lambda = 0$$

$$\boxed{\lambda = -2 + \frac{D_{12}}{D_{13}} + \frac{D_{12}}{D_{23}}}$$

$$\underline{A} = \begin{bmatrix} \left(-1 + \frac{D_{12}}{D_{13}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(\frac{-D_{12}}{D_{21}} + \frac{D_{12}}{D_{23}}\right) \end{bmatrix}$$

Projection matrices:

$$\rightarrow \quad adj(\lambda_1 \underline{I} - \underline{A}) = adj(-\underline{A}) = \begin{bmatrix} \left(+\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(+1 - \frac{D_{12}}{D_{13}}\right) \end{bmatrix}$$

$$S_1 = \frac{1}{0 - \lambda_2} adj(-\underline{A}) = \frac{1}{\left(2 - \frac{D_{12}}{D_{13}} - \frac{D_{12}}{D_{23}}\right)} \begin{bmatrix} \left(+\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(+1 - \frac{D_{12}}{D_{13}}\right) \end{bmatrix}$$

$$\rightarrow \quad (\lambda_2 \underline{I} - \underline{A}) = \begin{bmatrix} -2 + \cancel{\frac{D_{12}}{D_{13}}} + \frac{D_{12}}{D_{23}} + 1 - \cancel{\frac{D_{12}}{D_{13}}} & -\left(1 - \frac{D_{12}}{D_{13}}\right) \\ -\left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & -2 + \frac{D_{12}}{D_{13}} + \cancel{\frac{D_{12}}{D_{23}}} + \frac{D_{12}}{D_{21}} - \cancel{\frac{D_{12}}{D_{23}}} \end{bmatrix}$$

$$; \quad adj(\lambda_2 \underline{I} - \underline{A}) = \begin{bmatrix} -2 + \frac{D_{12}}{D_{13}} + \frac{D_{12}}{D_{21}} & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}} & -1 + \frac{D_{12}}{D_{23}} \end{bmatrix}$$

$$S_2 = \frac{1}{\left(-2+\frac{D_{12}}{D_{13}}+\frac{D_{12}}{D_{23}}\right)}\begin{bmatrix} -2+\frac{D_{12}}{D_{13}}+\frac{D_{12}}{D_{21}} & \left(1-\frac{D_{12}}{D_{13}}\right) \\ \frac{D_{12}}{D_{21}}-\frac{D_{12}}{D_{23}} & -1+\frac{D_{12}}{D_{23}} \end{bmatrix}$$

$$S_1+S_2 = I \;;\quad \frac{1}{\left(2-\frac{D_{12}}{D_{13}}-\frac{D_{12}}{D_{23}}\right)}\left(\begin{bmatrix} \cancel{\frac{D_{12}}{D_{21}}}-\frac{D_{12}}{D_{13}}+2-\frac{D_{12}}{D_{13}}-\cancel{\frac{D_{12}}{D_{21}}} & \cancel{1}-\cancel{\frac{D_{12}}{D_{13}}}-\cancel{1}+\cancel{\frac{D_{12}}{D_{13}}} \\ \cancel{\frac{D_{12}}{D_{21}}}-\cancel{\frac{D_{12}}{D_{23}}}-\cancel{\frac{D_{12}}{D_{21}}}+\cancel{\frac{D_{12}}{D_{23}}} & 1-\frac{D_{12}}{D_{13}}+1-\frac{D_{12}}{D_{23}} \end{bmatrix}\right)$$

$$\Rightarrow \frac{1}{\left(2-\cancel{\frac{D_{12}}{D_{13}}}-\cancel{\frac{D_{12}}{D_{23}}}\right)}\left(\begin{bmatrix} \left(2-\cancel{\frac{D_{12}}{D_{23}}}-\frac{D_{12}}{D_{13}}\right) & 0 \\ 0 & \left(2-\cancel{\frac{D_{12}}{D_{13}}}-\frac{D_{12}}{D_{23}}\right) \end{bmatrix}\right)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \underline{\underline{I}} \quad \checkmark$$

(viii) According to Sylvester's Theorem: $f(A) = \sum_{i=1}^{k} f(\lambda_i) A_i \quad ; \quad A_i = \prod_{j=1}^{k} \frac{1}{\lambda_i - \lambda_j}(A-\lambda_j I)$

$$f(A) = f(\lambda_1) A_1 + f(\lambda_2) A_2 \qquad A_1 = \frac{1}{\lambda_1-\lambda_2}(A-\lambda_2 I) = S_1$$

$$A_2 = \frac{1}{\lambda_2-\lambda_1}(A-\lambda_1 I) = S_2$$

$$@\ t=1 : \quad \underline{x}(t=1) = e^{n\phi^2 \underline{\underline{A}} t}\,\underline{x}_i + \underline{\underline{A}}^{-1}\left(e^{n\phi^2 \underline{\underline{A}} t} - \underline{\underline{I}}\right)\underline{b}$$

Evaluating the exponentials: $f(\underline{\underline{A}}) = e^{\underline{\underline{A}}}$

$$\therefore\ e^{\underline{\underline{A}}} = e^{\lambda_1} A_1 + e^{\lambda_2} A_2$$
$$= e^{\lambda_1} S_1 + e^{\lambda_2} S_2$$

$$\therefore\ \underline{x}(1) = e^{n\phi^2}\left(e^{\lambda_1} S_1 + e^{\lambda_2} S_2\right)\underline{x}_i + \underline{\underline{A}}^{-1}\left(e^{n\phi^2}\cdot\left(e^{\lambda_1} S_1 + e^{\lambda_2} S_2\right) - \underline{\underline{I}}\right)\underline{b} \quad ;\ \lambda_i = 0$$

$$= e^{n\phi^2} S_1 \underline{x}_i + e^{n\phi^2 \lambda_2} S_2 \underline{x}_i + \underline{\underline{A}}^{-1}\left(e^{n\phi^2} S_1 + e^{n\phi^2 \lambda_2} S_2 - \underline{\underline{I}}\right)\underline{b} \quad ;\ \underline{\underline{A}}^{-1} = \frac{1}{|A|}\,\mathrm{adj}(A)$$

$$= e^{n\phi^2} S_1 \underline{x}_i + e^{n\phi^2 \lambda_2} S_2 \underline{x}_i + \underline{\underline{A}}^{-1} e^{n\phi^2} S_1 \underline{b} + \underline{\underline{A}}^{-1} e^{n\phi^2 \lambda_2} S_2 \underline{b} - \underline{\underline{A}}^{-1} b$$

$$=$$

(ix) $$\eta = \frac{(e_1^T x_s)(e_2^T x_s)}{(e_1^T x_f)(e_2^T x_f)} = \frac{\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix} x_{1s} \\ x_{2s}\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix} x_{1s} \\ x_{2s}\end{bmatrix}}{\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix} x_{1s} \\ x_{2f}\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix} x_{1f} \\ x_{2f}\end{bmatrix}} = \frac{x_{1s}\cdot x_{2s}}{x_{1f}\cdot x_{2f}}$$

$$S_1 = \frac{1}{\left(2-\frac{D_{12}}{D_{13}}-\frac{D_{12}}{D_{23}}\right)}\begin{bmatrix}\left(+\frac{D_{12}}{D_{21}}-\frac{D_{12}}{D_{23}}\right) & \left(1-\frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}}-\frac{D_{12}}{D_{23}}\right) & \left(+1-\frac{D_{12}}{D_{13}}\right)\end{bmatrix} = \frac{1}{\left(2-\frac{3}{2}-\frac{3}{2}\right)}\begin{bmatrix} 1-\frac{3}{2} & 1-\frac{3}{2} \\ 1-\frac{3}{2} & 1-\frac{3}{2}\end{bmatrix} = -1\begin{bmatrix} -1/2 & -1/2 \\ -1/2 & -1/2\end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2\end{bmatrix}$$

$$S_2 = \frac{1}{\left(-2+\frac{D_{12}}{D_{13}}+\frac{D_{12}}{D_{23}}\right)}\begin{bmatrix} -2+\frac{D_{12}}{D_{13}}+\frac{D_{12}}{D_{21}} & \left(1-\frac{D_{12}}{D_{13}}\right) \\ \frac{D_{12}}{D_{21}}-\frac{D_{12}}{D_{23}} & -1+\frac{D_{12}}{D_{23}}\end{bmatrix} = \frac{1}{\left(-2+\frac{3}{2}+\frac{3}{2}\right)}\begin{bmatrix} -2+\frac{3}{2}+1 & 1-\frac{3}{2} \\ 1-\frac{3}{2} & -1+\frac{3}{2}\end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2\end{bmatrix}$$

$$\underline{b} = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix} \qquad \therefore \lambda_2 = -2 + \frac{3}{2} + \frac{3}{2} = 1$$

$$\underline{x}_s = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} x_{1f} \\ x_{2f} \end{bmatrix} + e^{+n\phi^2} \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} x_{1f} \\ x_{2f} \end{bmatrix} + \eta\phi^2 \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix} + \frac{1}{1} \begin{bmatrix} e^{n\phi^2} - 1 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix}$$

↳ Solving for x_{1s} and x_{2s} from this expression to get in terms of x_{1f}, x_{2f}

$$\underline{x}_s = \begin{bmatrix} \frac{1}{2}x_{1f} + \frac{1}{2}x_{2f} \\ \frac{1}{2}x_{1f} + \frac{1}{2}x_{2f} \end{bmatrix} + e^{n\phi^2} \begin{bmatrix} \frac{1}{2}x_{1f} - \frac{1}{2}x_{2f} \\ -\frac{1}{2}x_{1f} + \frac{1}{2}x_{2f} \end{bmatrix} + n\phi^2 \begin{bmatrix} 6/4 \\ 6/4 \end{bmatrix}$$

$$\underline{x}_s = \begin{bmatrix} \left(\frac{1}{2} + e^{n\phi^2}\frac{1}{2}\right)x_{1f} + \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right)x_{2f} \\ \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right)x_{1f} + \left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right)x_{2f} \end{bmatrix} + n\phi^2 \begin{bmatrix} 6/4 \\ 6/4 \end{bmatrix}$$

$$\underline{x}_s = \begin{bmatrix} \left(\frac{1}{2} + e^{n\phi^2}\frac{1}{2}\right) & \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right) \\ \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right) & \left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right) \end{bmatrix} \underbrace{\begin{bmatrix} x_{1f} \\ x_{2f} \end{bmatrix}}_{\underline{x}_f} + n\phi^2 \begin{bmatrix} 6/4 \\ 6/4 \end{bmatrix}$$

$$\eta = \frac{(e_1^T \underline{x}_s)(e_2^T \underline{x}_s)}{(e_1^T \underline{x}_f)(e_2^T \underline{x}_f)} = \frac{\left(\left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right)x_{1f} + \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right)x_{2f} + n\phi^2 6/4\right)\left(\left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right)x_{1f} + \left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right)x_{2f} + \phi^2 n\frac{6}{4}\right)}{x_{1f}\, x_{2f}}$$

$$= \frac{x_{1f}^2\left(\frac{1}{4} - \frac{e^{2n\phi^2}}{4}\right) + x_{2f}^2\left(\frac{1}{4} - \frac{e^{2n\phi^2}}{2}\right) + x_{1f}x_{2f}\left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right)^2 + x_{1f}x_{2f}\left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right)^2 + x_{1f}\phi^2 n\frac{6}{4} + x_{2f}}{x_{1f}\, x_{2f}}$$

(x) Plugging in $x_{1f} = x_{2f}$

↳ will give the required expression